

Self-Referential Structures with Pointers (Self-Learning)

Q. Explain the concept of Structures in C. How can we use pointers with structures? Provide a code example to demonstrate its use.

> **Definition to Remember**

> A **Structure** is a user-defined data type that can contain one or more variables of different data types. It is used for grouping related data items under a single name. Structures are passed to functions by value or by reference (using pointers).

1. Structures in C

- Groups multiple variables (called **members/fields**) of different types into one logical unit.

- Ide

2. Pointers with Structures

When a structure is passed to a function **by value**, a full copy is made, which is wasteful for large structures. Passing a **pointer** to the structure is more efficient.

- **Arrow Operator** (`->`) is used to access members via a pointer.

- `ptr`

3. Self-Referential Structure

A structure that contains a **pointer to another structure of the same type** is called a self-referential structure. It is the building block of **Linked Lists and Trees**.

```c

### 4. Complete Code Example

```c

#include

struct Student {

int roll_

```
int main() {  
  
    // Dynamic Memory Allocation  
  
    if (ptr == NULL) {  
  
        printf("Memory not allocated\n");  
  
        return 1;  
    }  
  
    // Access members using Arrow Operator (->)  
  
    ptr->name = "Suraj";  
  
    printf("Student Name: %s\n", ptr->name);  
  
    free(ptr); // Release allocated memory  
  
    return 0;  
}
```

5. Dot vs Arrow Operator

| Situation | Operator | Syntax |

```
---  
> **Must-Write Points (for 10 marks)**  
  
---  
> 1. `s`
```

```
---  
> **Quick Recall**  
  
---  
> `stru`  
to rele
```

C Code: Self-Referential Structure Node Creation & Linked List Traversal

```
``c  
#include <stdio.h>
```

```
// Definition of Self-Referential Structure
```

```
struct l
```

```
// Function to traverse dynamic linked list
```

```
void pr
```

```
int main() {
```

```
// Dyna
```

```
// Assigning data & linking nodes
```

```
head->
```

```
second->data = 20;
```

```
secon
```

```
third->data = 30;
```

```
third->
```

```
printList(head);
```

```
// Free allocated memory
```

```
free(h
```

```
return 0;
```

```
}
```